

V3 moneyness performance

Seven models · four companies · 4 windows · 1.5-year fixed calibration · classified by call moneyness $m = S/K$

What this report is

- Companion to V4_1p5y_fixed_empirical_study. The notebook in this folder is the source of truth; this PDF is written from the same moneyness payload.
- Question: does model ranking change with option moneyness, overall and inside each volatility regime?
- BEST in every table is the lowest RMSE%. That row is bold and green, and the Mark column says BEST. Empty buckets ($n = 0$) show em dashes and have no BEST.

Unchanged experimental design

- Same seven models, four companies (SPY, AAPL, MSFT, AMZN), 4 windows, 1.5-year lookback, none recalibration, V3 filters, Euler-corrected Heston step, LSM $n_{\text{paths}} = 10000$ / seed 42.
- Same option contracts as the main study (nearest-ATM listed calls, DTE 7–60, $|S/K - 1| \leq 10\%$). No model-specific re-sampling.
- Pooled $n = 802$ contracts per model (identical keys across models).

Moneyness rule (calls, $m = S / K$)

- DOTM: $m < 0.98$. OTM: $0.98 \leq m < 0.995$. ATM: $0.995 \leq m \leq 1.005$. ITM: $1.005 < m \leq 1.02$. DITM: $m > 1.02$.
- These five bins partition the main-study filter $|m - 1| \leq 10\%$. The Monday nearest-ATM design puts most mass in ATM; DOTM and DITM are sparse and appear mainly in 2008–2009.

Metrics

- $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})^2)}$. Ranking uses this only.
- MAE and Bias \$ in option-premium dollars. $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Early-exercise fraction = mean LSM share of paths that stop before expiry.

Two analyses

- Analysis 1 — Overall moneyness: pool all companies and regimes. Five tables (one per bucket) plus a Model $\times$ Moneyness RMSE% heatmap.
- Analysis 2 — Window $\times$ moneyness: keep the 4 windows separate, pool the four companies. Compact RMSE tables/heatmaps and a winner grid showing whether the best model changes with moneyness inside each window.

Table M0 · Contract counts (one model = all models)

Regime	DOTM	OTM	ATM	ITM	DITM	Total
2008-2009 Crisis	18	43	80	43	16	200
2013-2014 Normal	0	5	192	3	0	200
2018-2019 Late-cycle	0	4	191	7	0	202
2019-2020 COVID	0	6	189	5	0	200
All regimes	18	58	652	58	16	802

Moneyness rule · American calls, $m = S/K$

Bucket	Rule	n (pooled)	Share
DOTM	$m < 0.98$	18	2.2%
OTM	$0.98 \leq m < 0.995$	58	7.2%
ATM	$0.995 \leq m \leq 1.005$	652	81.3%
ITM	$1.005 < m \leq 1.02$	58	7.2%
DITM	$m > 1.02$	16	2.0%

ATM dominates because the main study picks the nearest-ATM listed call each Monday. DOTM/DITM are not empty because of a new filter — they are rare in that ATM sample, especially after 2009. Every model still sees the same contracts in every bucket.

Page 3 · Analysis 1 · Overall moneyness · Tables M1-M3

Pooled SPY + AAPL + MSFT and all 4 windows. Bold green = lowest RMSE%.

Table M1 · DOTM · $m < 0.98$ · $n = 18$ · BEST = Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	18	34.14	1.2323	-0.4208	-3.15	18.8%	6
Modified GBM	18	33.14	1.2868	-0.7616	-11.27	19.8%	4
GARCH	18	33.76	1.2330	-0.4053	-0.45	18.6%	5
Heston	18	30.92	1.2195	-0.8021	-12.12	15.1%	3
Merton	18	30.69	1.1924	-0.6232	-8.16	12.2%	1
GARCH-Merton	18	53.89	1.4954	+0.6023	+25.64	14.1%	7
Heston-Merton	18	30.77	1.2060	-0.6403	-8.44	9.1%	2

Table M2 · OTM · $0.98 \leq m < 0.995$ · $n = 58$ · BEST = Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	58	43.05	1.2320	-0.0840	+5.78	21.7%	5
Modified GBM	58	40.29	1.1889	-0.3358	+0.62	22.7%	2
GARCH	58	41.78	1.1965	-0.0659	+10.40	20.3%	3
Heston	58	42.66	1.1901	-0.2036	+4.98	15.8%	4
Merton	58	40.28	1.1820	-0.1032	+5.58	17.2%	1
GARCH-Merton	58	59.12	1.4946	+0.6796	+31.70	16.0%	7
Heston-Merton	58	45.60	1.2180	+0.0719	+11.01	11.2%	6

Table M3 · ATM · $0.995 \leq m \leq 1.005$ · $n = 652$ · BEST = GARCH

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	652	41.64	4.4486	+0.8708	+14.16	24.0%	4
Modified GBM	652	41.49	4.5182	+1.1699	+12.76	24.5%	3
GARCH	652	38.49	4.0841	-0.6047	+10.28	22.4%	1
Heston	652	44.79	4.5911	+0.8212	+19.18	17.4%	5
Merton	652	41.46	4.5326	+0.8942	+14.34	18.8%	2
GARCH-Merton	652	51.30	4.6488	+1.5259	+26.29	18.4%	6
Heston-Merton	652	56.15	5.8676	+3.5368	+35.03	11.8%	7

Bias \$ = mean(model – market). Bias% = 100 × mean((model – market)/market).

Page 4 · Analysis 1 · Overall moneyness · Tables M4-M5

Pooled SPY + AAPL + MSFT and all 4 windows. Bold green = lowest RMSE%.

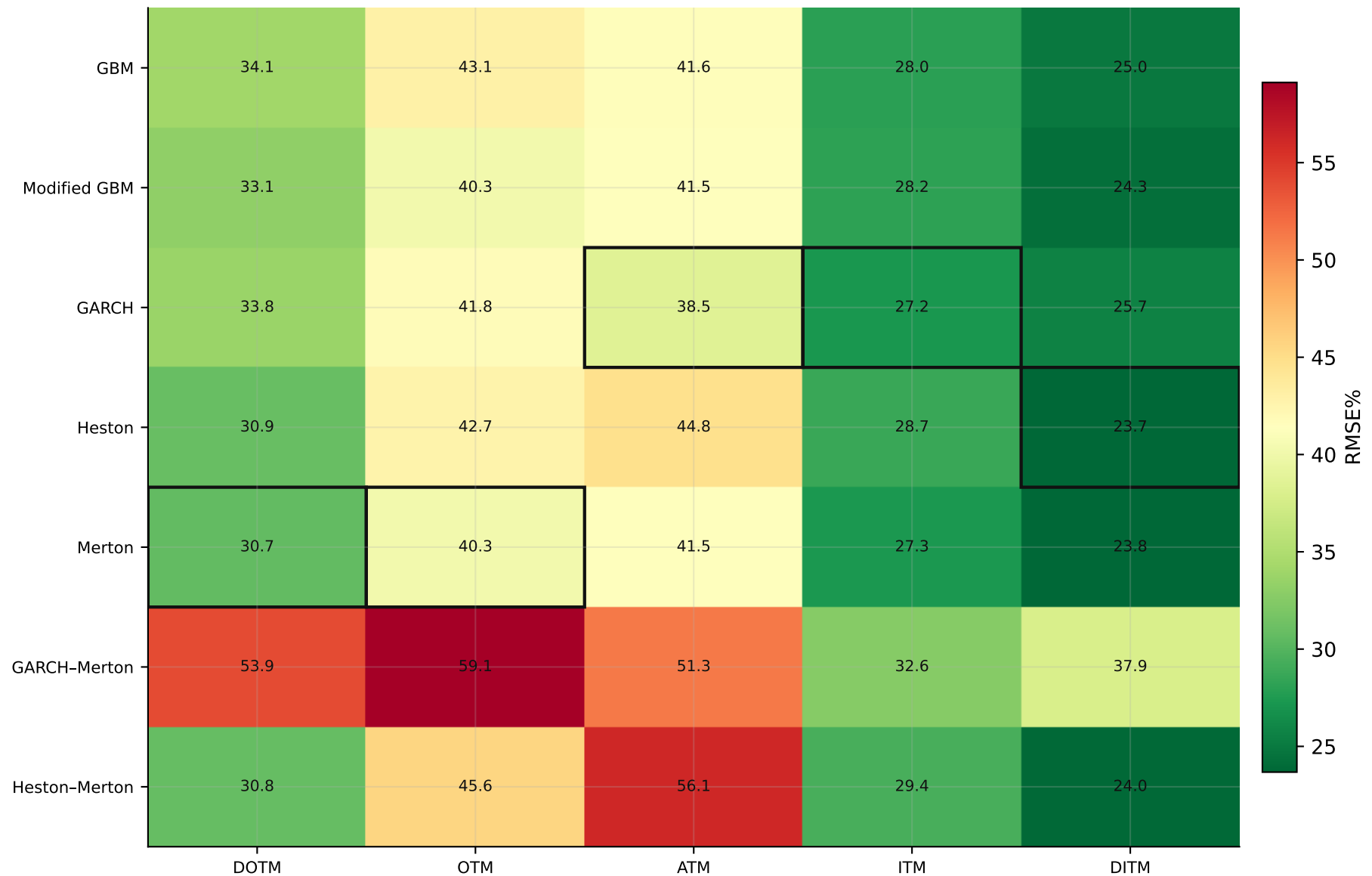
Table M4 · ITM · $1.005 < m \leq 1.02$ · $n = 58$ · BEST = GARCH

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	58	28.03	1.0069	-0.4877	-10.20	27.6%	3
Modified GBM	58	28.24	0.9858	-0.6538	-12.90	28.7%	4
GARCH	58	27.17	0.9648	-0.4008	-3.60	25.6%	1
Heston	58	28.72	1.0304	-0.5118	-9.09	21.3%	5
Merton	58	27.33	1.0220	-0.3921	-6.74	20.3%	2
GARCH-Merton	58	32.59	1.1342	+0.2247	+10.23	20.0%	7
Heston-Merton	58	29.45	1.0780	-0.1324	-1.56	13.5%	6

Table M5 · DITM · $m > 1.02$ · $n = 16$ · BEST = Heston

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Rank
GBM	16	25.04	1.2933	-0.3293	-3.90	29.2%	5
Modified GBM	16	24.31	1.2710	-0.6582	-9.33	30.8%	4
GARCH	16	25.66	1.3065	-0.2992	-0.57	27.2%	6
Heston	16	23.68	1.2301	-0.7548	-10.55	22.0%	1
Merton	16	23.80	1.2447	-0.4629	-4.98	24.5%	2
GARCH-Merton	16	37.87	1.4533	+0.7255	+18.85	20.8%	7
Heston-Merton	16	23.95	1.2472	-0.3542	-2.80	13.3%	3

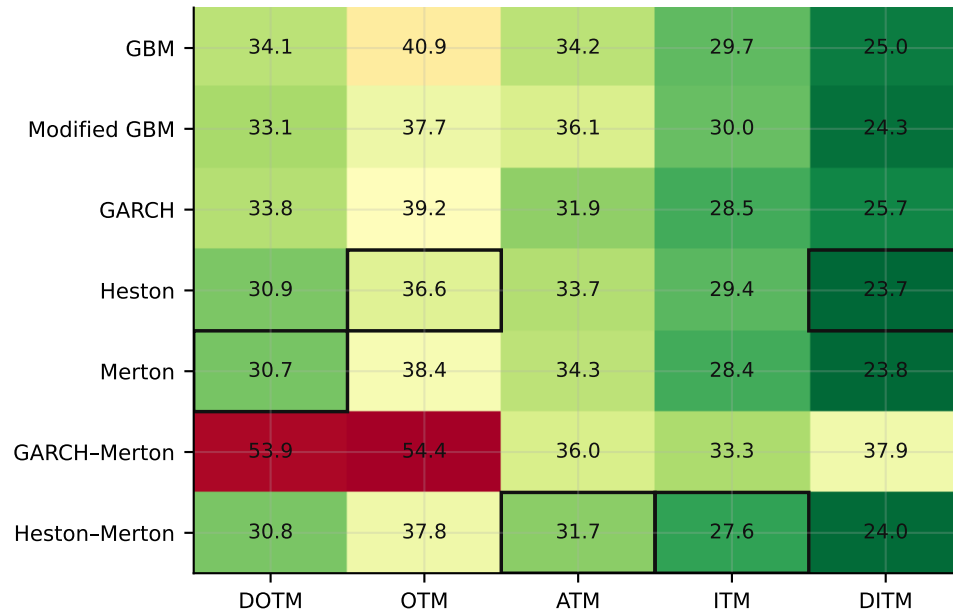
ITM/DITM: $S > K$. DOTM/OTM: $S < K$. ATM: $|S/K - 1| \leq 0.5\%$.



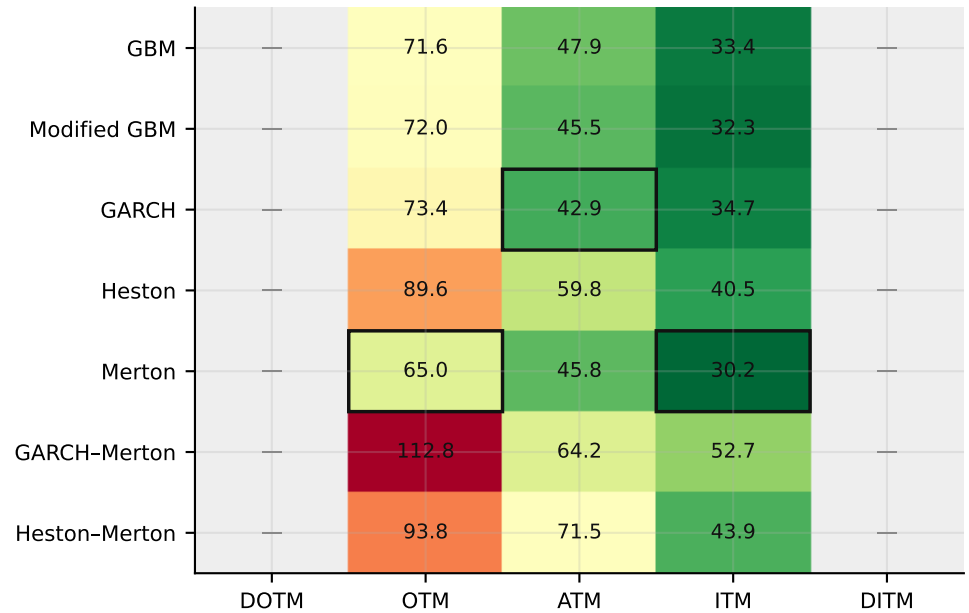
Black outline = lowest RMSE% in that moneyness column. Grey = n = 0. Pooled across companies and regimes.

Figure M2 · RMSE% by model × moneyness · each window (companies pooled)

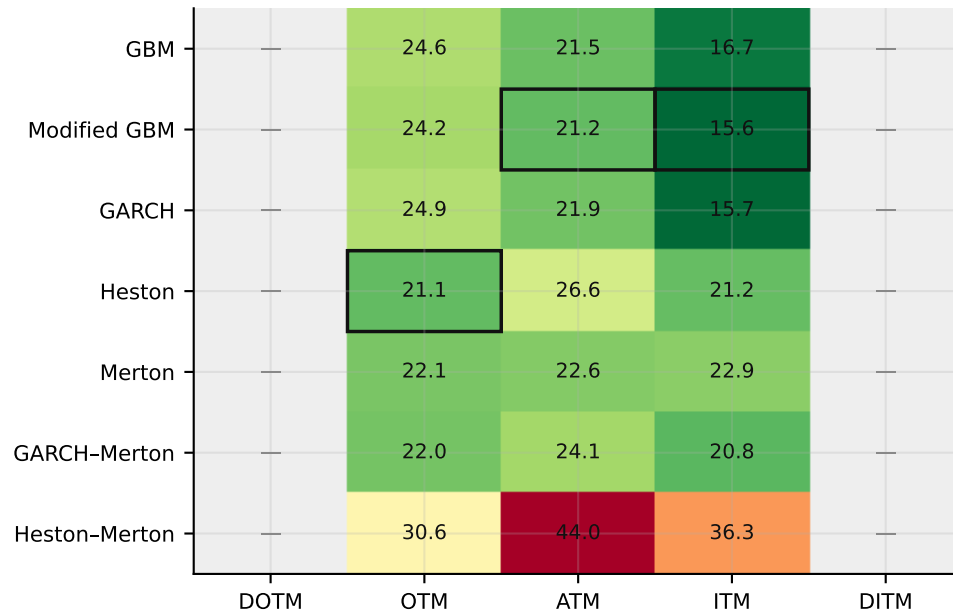
2008-2009 · Crisis



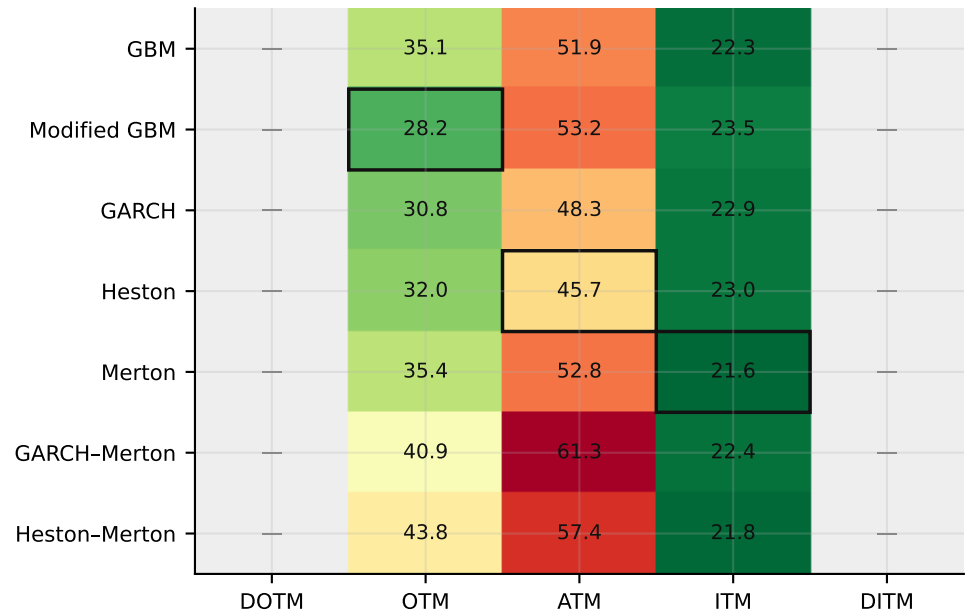
2013-2014 · Normal



2018-2019 · Late-cycle



2019-2020 · COVID



Outlined cell = lowest RMSE% in that window × moneyness column. Grey = n = 0.

Analysis 2 · Does the ranking change with moneyness inside each window?

Table M6 · Best model (lowest RMSE%) by window × moneyness

Window	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	Merton	Heston	Heston–Merton	Heston–Merton	Heston
2013-2014 Normal	—	Merton	GARCH	Merton	—
2018-2019 Late-cycle	—	Heston	Modified GBM	Modified GBM	—
2019-2020 COVID	—	Modified GBM	Heston	Merton	—

Table M7 · n contracts by window × moneyness (shared across models)

Window	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	18	43	80	43	16
2013-2014 Normal	0	5	192	3	0
2018-2019 Late-cycle	0	4	191	7	0
2019-2020 COVID	0	6	189	5	0

A change in the Table M6 entry across a row means the ranking depends on moneyness inside that window. Columns with n = 0 cannot change the ranking.

Does performance change with moneyiness?

- DOTM (n = 18): BEST = Merton with RMSE% = 30.69, Bias% = -8.16.
- OTM (n = 58): BEST = Merton with RMSE% = 40.28, Bias% = +5.58.
- ATM (n = 652): BEST = GARCH with RMSE% = 38.49, Bias% = +10.28.
- ITM (n = 58): BEST = GARCH with RMSE% = 27.17, Bias% = -3.60.
- DITM (n = 16): BEST = Heston with RMSE% = 23.68, Bias% = -10.55.
- Overall, the ranking does change with moneyiness: DOTM → Merton, OTM → Merton, ATM → GARCH, ITM → GARCH, DITM → Heston.
- Inside regimes, the BEST model also changes with moneyiness in: 2008-2009, 2013-2014, 2018-2019, 2019-2020.
- These comparisons use the Monday nearest-ATM sample, so ATM is the only well-populated bucket after 2009. DOTM/DITM results rest on few 2008–2009 quotes and should be read as descriptive, not as a high-powered test.